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G.Chandra sekhar Reddy
192472173

Analytical Problem Solving ITA0502 CV

① Grey level Transformation

(a) purpose:

grey level transformation modifies pixel intensities to improve image quality. it is used to increase brightness, improve contrast, highlight image details, or create image negatives for better visualization

(b) Negative Transformation

formula:

$s = 255 - r$

original pixel(r)	Transformed pixel(s)
50	205
100	155
150	105
200	55

Interpretation: Dark pixels become bright and bright pixels become dark, producing a negative image that enhances hidden details.

② Histogram Equalization

(a) purpose

Histogram equalization improves image contrast by spreading pixel intensities over the available range, making details more visible.

(b) Cumulative Distribution Function (CDF)

Intensity	Frequency	CDF
0	4	4
1	6	10
2	10	20
3	6	26

Total pixels = 26

(c). Equalized Intensities

Using:-

$$s = \frac{CDF}{26} \times 3$$

Intensity	Equalized value
0	0
1	1
2	2
3	3

Interpretation: The pixel values become more evenly distributed, resulting in better image contrast

(3) Gaussian Smoothing

(a) why Gaussian filter?

Gaussian filtering reduces noise while preserving edges better than an averaging filter. It gives more weight to the center pixel, producing smoother and more natural result.

(b) Calculation

Kernel:- $\frac{1}{16}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$

C. Chandra Sekhar Reddy

Image:- $\begin{bmatrix} 10 & 20 & 10 \\ 20 & 40 & 20 \\ 10 & 20 & 10 \end{bmatrix}$

weighted sum:-

$= (10+40+10+40+160+40+10+40+10)/16$

$= 360/16 = 22.5 \approx 23$

Interpretation:- The center pixel changes from 40 to about 23, reducing noise and smoothing the image.

(4) Edge Detection using Sobel operator

(a) purpose

Edge detection identifies boundaries where image intensity changes rapidly. it is useful for object detection and image segmentation.

(b) Sobel Calculation

mask :- $\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$

Image:- $\begin{bmatrix} 10 & 10 & 10 \\ 20 & 20 & 20 \\ 30 & 30 & 30 \end{bmatrix}$

gradient:-

$= (-10-20-10) + (30+60+30)$

$= -40 + 120$

$= 80$

Interpretation; A gradient of 80 indicates a strong horizontal edge at the center pixel.

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(5) Contrast stretching

(a) purpose

Contrast stretching expands a narrow intensity range to the full display range, making dark regions darker, bright regions brighter, and improving overall image clarity.

(b) Linear Contrast stretching

Formula:

$$s = \frac{(r-50) \times 255}{150-50}$$

for $r=50$:

$s = 0$

for $r=100$:

$$s = \frac{50 \times 255}{100} = 127.5 \simeq 128$$

For $r=150$:

$s = 255$

original value	New value
50	0
100	128
150	255

Interpretation: The image now uses the full intensity range (0-255), giving higher contrast and clearer details.